# Fusing Frequency-Domain Features and Brain Connectivity Features for Cross-Subject Emotion Recognition

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***Abstract*— Frequency-domain (FD) features reveal the activated patterns of individual local brain regions responding to different emotions, whereas brain connectivity (BC) features involve the coordination of multiple brain regions for generating emotional responses; these two types of features are complementary to each other. To date, the fusion of these two types of features for electroencephalography (EEG)-based cross-subject emotion recognition remains to be fully investigated due to the intersubject variability in EEG signals. In this article, we first attempt to investigate these fused features for cross-subject emotion recognition from multiple perspectives, including critical frequency bands, complementary characteristics for each emotional state, critical channels, and crucial connections, using a fast and robust approximate empirical kernel map-fusion-based support vector machine (AEKM-Fusion-SVM) method. The experimental results on the SJTU emotion EEG dataset (SEED), BCI2020-A, and BCI2020-B datasets reveal that: 1) the AEKM-fusion method improves the effectiveness and efficiency of the fusion of features of different dimensions; 2) the recognition accuracy of the fused features outperforms each individual feature, and this outperformance is more significant in the high-frequency bands (i.e., the beta and gamma bands); 3) the fused features significantly enhance the classification performance for negative emotion; and 4) the fused features built with 27 selected channels achieve comparable performance to that of the fused features built with the full number of channels (i.e., 62 channels), allowing for easier establishment of brain–computer interface (BCI) systems in real-world scenarios. Our study enriches the research of emotion-related brain mechanisms and provides new insight into affective computing.**

***Index Terms*— Approximate empirical kernel map (AEKM), brain connectivity (BC) features, cross-subject emotion recognition, frequency-domain (FD) features, fused features.**

Manuscript received November 22, 2021; revised March 8, 2022; accepted March 30, 2022. Date of publication April 20, 2022; date of current version May 3, 2022. This work was supported in part by the Guangdong Basic and Applied Basic Research Foundation under Grant 2020A1515111154; in part by the Special Projects in Key Fields Supported by the Technology Development Project of Guangdong Province under Grant 2020ZDZX3018; in part by the Special Fund for Science and Technology of Guangdong Province under Grant 2020182; in part by the Educational Commission of Guangdong Province under Grant 2021KTSCX136; in part by the Wuyi University and Hong Kong and Macao Joint Research Project under Grant 2019WGALH16; in part by the Jiangmen Science and Technology Project under Grant 2021030100050004285; in part by the European Union's Horizon 2020 Research and Innovation Program [Proactive Safety Systems and Tools for a Constantly Upgrading Road Environment (SAFE_UP)] under Grant 031019; and in part by the Science and Technology Development Fund, Macau, under Grant 0045/2019/AFJ. The Associate Editor coordinating the review process was Dr. David Ayllon. *(Chuangquan Chen and Zhencheng Li contributed equally to this work.) (Corresponding author: Hongtao Wang.)*

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Digital Object Identifier 10.1109/TIM.2022.3168927

## I. Introduction

EMOTION recognition is a popular topic for application in various fields, including the development of human–computer interaction systems, determination of the level of interest in learning, identification of the level of vigilance on roads for safety, detection of patients' mental and physical states, provision of medical care activities, and prevention of emotional disorders in children [1]–[5]. To date, a variety of approaches have been applied to recognize emotions, including speech, facial expressions, physiological measurement, and self-assessment [6]–[9]. Among these available indicators, physiological signals, including electrocardiography (ECG), blood pressure (BP), electromyography (EMG), and electroencephalography (EEG) signals, are beneficial in which they are difficult to mask or disguise [10]. In recent years, EEG-based emotion recognition has garnered increasing attention in both research and application [4], [11], [12], due in part to the rapid development of noninvasive, simple-to-use, and low-cost EEG recording devices.

EEG-based emotion recognition relies on the extraction of sufficiently discriminative EEG features. The extracted features from a multichannel EEG time series can be coarsely grouped into three categories: temporal-domain features, frequency-domain (FD) features, and brain connectivity (BC) features. Among these categories, FD features and BC features have been demonstrated to be the most accurate and reliable predictors for emotion recognition. In the FD, EEG signals are analyzed in terms of the classical delta, theta, alpha, beta, and gamma frequency bands. FD features, including the power spectral density (PSD) and differential entropy (DE) [13], are extracted from each EEG signal channel independently in specific frequency bands. Jiang *et al.* [14] investigated the properties of the PSD for cross-subject emotion recognition at different frequency bands. Duan *et al.* [15] proposed a novel DE feature for EEG-based emotion recognition, and Zheng *et al.* [13] and Zheng and Lu [16] employed DE fea-

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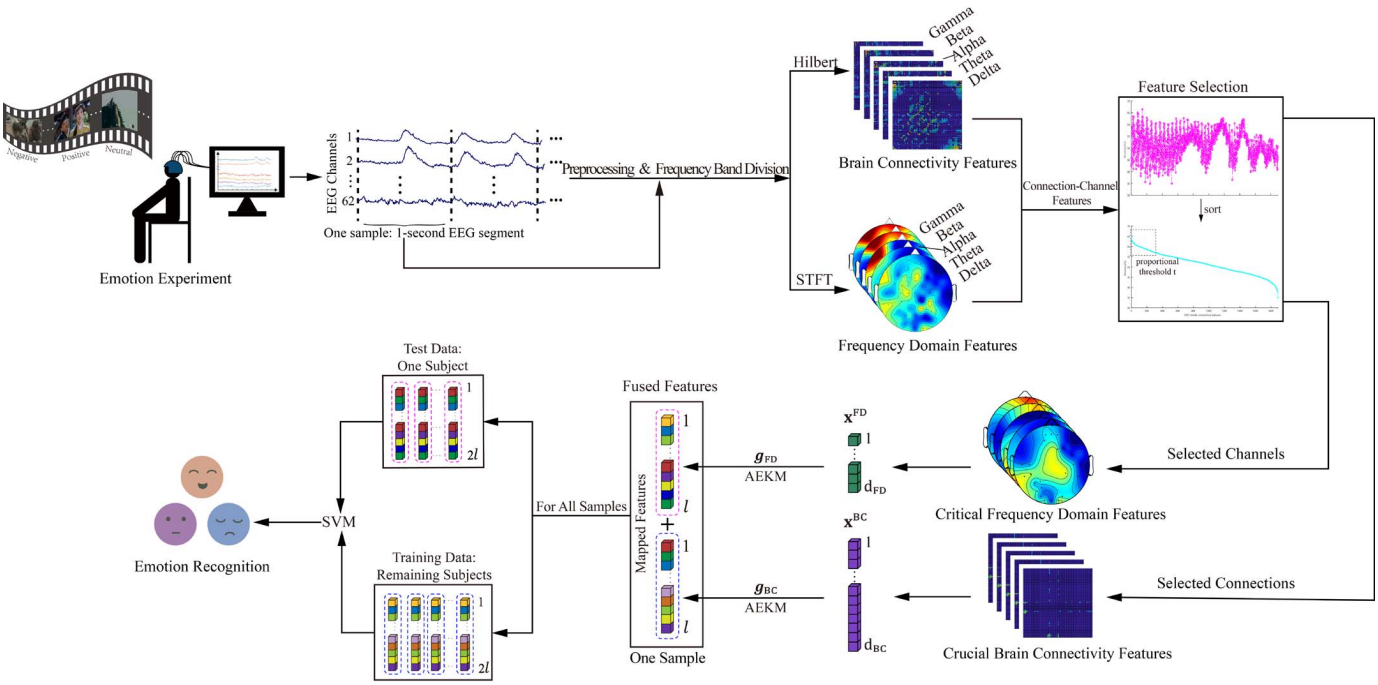



Fig. 1. Flowchart of cross-subject emotion recognition based on fused features. First, EEG data are collected as the subject is watching movie clips with different emotional labels. Second, the FD features and the BC features are extracted from the EEG data, and the connection-channel feature is constructed to select the crucial features proportionally. Finally, the selected FD features and BC features are mapped through AEKM and then concatenated and fed into the SVM to build a cross-subject emotion recognition model.

tures to identify the stable patterns over time and the critical frequency bands and channels for emotion recognition. However, since the FD features are local features that only offer information within a channel, they ignore the spatial information from EEG signals, making them unsuitable for exploring the interactions between multiple brain regions. In contrast, BC features, as spatial features, can capture spatial information by characterizing the relationship between channels. As two of the most commonly used BC features, the phase lag index (PLI) and phase lock value (PLV) [17], [18] quantify the phase synchronization among electrode signals across multiple brain regions. Gonuguntla *et al.* [19] demonstrated that the distinctive associated areas for emotions can be identified with the PLV feature. Liu *et al.* [20] adopted the PLI feature to identify the three emotional states of individual subjects and showed its discriminability.

Recent research has shown that combining different types of features can significantly increase the accuracy of EEG-based emotion recognition. Soleymani *et al.* [21] used the PSD features extracted from EEGs and collected eye-gaze data to conduct cross-subject emotion recognition experiments. Zheng *et al.* [22] directly concatenated the feature vectors extracted from EEG and eye movements and inputted them into a support vector machine (SVM) to classify different emotions. Liu *et al.* [23] employed a deep canonical correlation analysis (DCCA) of multimodal signals, resulting in the identification of more discriminative features across emotions. Li *et al.* [24] conducted feature-fusion research on DE features, PSD feature, and EEG-based network patterns (ENPs) (i.e., *C*-coefficient, SP-length, *G*-efficiency, and *L*-efficiency extracted from the PLV), investigating the classification accuracy of fused features regarding intrasubject emotion recognition.

However, to the best of our knowledge, no studies have examined the properties of fused features based on FD features (e.g., the PSD and DE) and BC features (e.g., the PLV and PLI) for cross-subject emotion recognition. FD features offer information within a channel, whereas BC features characterize the relationship between channels; thus, these two sets of features are complementary to each other. We believe that the fusion of these two types of features will improve the classification accuracy of cross-subject emotion recognition.

Therefore, in this article, we first attempt to investigate the properties of these fused features for cross-subject emotion recognition using a fast and robust approximate empirical kernel map-fusion-based SVM (AEKM-fusion-SVM) method. In particular, each set of features is first transformed into the same dimensions with Nyström-based AEKM [25]–[29], which are then concatenated and fed into an SVM classifier. We employ leave-one-subject-out cross validation to assess the performance of the fused features. The flowchart of cross-subject emotion recognition based on the fused features is shown in Fig. 1. Our work significantly differs from Li *et al.*'s [24] work in the following aspects.

1) Unlike Li *et al.*'s [24] work, which employed the *C*-coefficient, SP-length, *G*-efficiency, and *L*-efficiency extracted from the PLV for fusion with the FD features, we directly adopt the original BC features (i.e., the PLV and PLI) for fusion. We leverage the AEKM-fusion method to effectively and efficiently fuse features of different dimensions.

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2) In contrast to Li *et al.*'s [24] work, which investigated the classification accuracy of fused features on intrasubject emotion recognition, we apply the fused features to cross-subject emotion recognition and further investigate the properties of the fused features in-depth.

The main contributions of this article are given as follows.

1) We directly adopt the original BC features for fusion and leverage the fast and robust AEKM-fusion method to improve the effectiveness and efficiency of the fusion of features of different dimensions.

2) We verify that the recognition accuracy of the fused features across subjects outperforms each individual feature. In particular, the advantages of classification in the high-frequency bands (i.e., the beta and gamma bands) are more obvious.

3) We reveal the complementary characteristics of the fused features. After combining FD features and BC features and feeding them into the SVM, the classification performance for negative emotions is significantly improved.

4) We develop a novel perspective that studies the critical channels and important connections for cross-subject emotion recognition in terms of the fused features.

The rest of this article is structured as follows. Section II briefly introduces the adopted materials and methodology. A comprehensive description of the methodologies, including preprocessing, feature extraction, feature smoothing, feature mapping, and the classifier, is provided in this section. Section III reports the results of the fused features for cross-subject emotion recognition, including feature exhibition, the classification performance of different fused features, and investigations of the properties of the fused features. Section IV presents a discussion of the results from Section III. Finally, the conclusions are summarized in Section V.

## II. Materials and Methodologies

In this section, we introduce the datasets and methods employed in this study.

### A. Datasets

In this article, we adopt three different emotion datasets (i.e., the SJTU emotion EEG dataset (SEED), BCI2020-A, and BCI2020-B) to investigate the properties of fused features for cross-subject emotion recognition. SEED is a widely adopted public emotion recognition dataset [16], while BCI2020-A and BCI2020-B are the datasets employed in the 2020 World Robot Contest-brain–computer interface (BCI) Controlled Robot Contest for cross-subject emotion recognition, provided by Shanghai Jiao Tong University. Since the experimental paradigms of these three emotional datasets are the same, differing only in the number of subjects (i.e., 15 subjects in the SEED dataset and 23 subjects in the other two datasets), we briefly introduce only the SEED dataset here.

During the experiment, the subjects were required to watch movie clips labeled with one of three different emotions, i.e., negative, positive, and neutral, and each movie clip was approximately 4 min long. Each experiment had a total of 15 trials. There was a 5-s hint before each clip, after which the subjects were given 45 s for self-assessment and 15 s for rest. The sequence of presentations was arranged such that two movie clips with the same emotions were not shown consecutively. Participants were instructed to complete a questionnaire with scores of 1–5 and keywords immediately after watching each clip to provide feedback about their positive, neutral, and negative emotions. Fig. 2 shows the detailed protocol. As shown in Fig. 3 [15], [16], an EEG was recorded using an Neuro-Scan System at a sampling rate of 1000 Hz with a 62-channel active AgCl electrode cap according to the international 10–20 system.

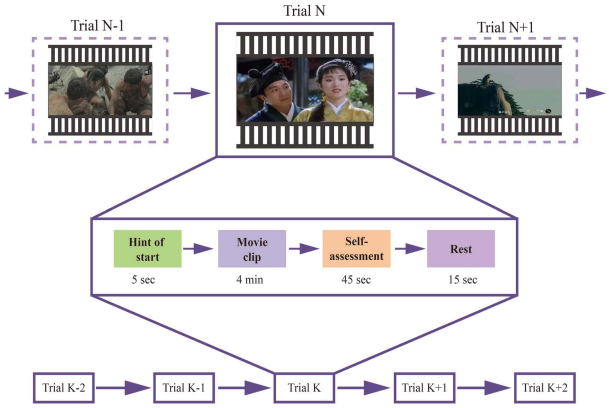



Fig. 2. Data collection process in the experiment.

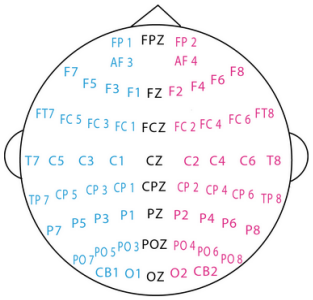



Fig. 3. Sixty two-channel EEG cap.

### B. Preprocessing

Following [13], [16], [30], and [31], the raw EEG signals are sliced into a series of 1-s segments without overlap. EEG signals have high temporal resolution, and 1 s is sufficient to identify human emotional states [30], [31]. In the preprocessing stage, the raw data are downsampled to 200 Hz to expedite the computation. To remove irrelevant artifacts, all raw EEG signals are filtered through a 1–50 Hz bandpass filter. Subsequently, for further exploration of the stable frequency bands in cross-subject emotion recognition, the signals are filtered with five passbands, namely, delta (1–4 Hz), theta (4–8 Hz), alpha (8–14 Hz), beta (14–31 Hz), and gamma (31–50 Hz).

### C. Feature Extraction

*1) FD Features:* They reflect the power differences in brain regions with respect to different emotions [24]. In this article,

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we only focus on the PSD and DE features since they have demonstrated superiority over other FD features [13].

Given a processed 1-s segment, we extract the PSD and DE features at the delta ($\delta$), theta ($\theta$), alpha ($\alpha$), beta ($\beta$), and gamma ($\gamma$) frequency bands. The traditional PSD features are extracted using a short-time Fourier transform (STFT) and a nonoverlapping Hanning window. EEG signals of a specific channel in an individual band are approximately subject to a Gaussian distribution $N(\mu, \sigma^2)$. For a given segment, the DE feature of each channel in each frequency band can be simply estimated as follows [32]:

$$\begin{aligned} \mathrm{DE} &= -\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{(x-\mu)^2}{2\sigma^2}} \log_2\left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{(x-\mu)^2}{2\sigma^2}}\right) dx \\ &= \frac{1}{2}\log_2\left(2\pi e\sigma^2\right) \end{aligned} \tag{1}$$

where $\pi$ and $e$ are constants. The DE is equivalent to the logarithmic PSD for a fixed-length EEG sequence in a specific frequency band according to [32].

Since each frequency band signal includes 62 channels, the DE and PSD features for a sample have 310 dimensions (62 channels $\times$ 5 bands).

*2) BC Features:* Emotional responses in the brain involve the coordination of multiple cerebral cortex regions. Therefore, the brain can be treated as a complex network with connectivity to explore the interactions between multiple brain regions. We utilize the PLV and PLI, two of the most commonly used connectivity features for EEG-based emotion recognition, to quantify the phase synchronization among electrode signals across multiple brain regions. The computation of the PLI and PLV is detailed as follows [33], [34].

Given a time series $s_i(t)(t = 1, 2, 3, \ldots, T)$ with $T$ sample points obtained from the $i$th EEG channel, the corresponding analytical signal $z_i(t)$ is complex-valued defined as

$$z_i(t) = s_i(t) + j\tilde{s}_i(t) \tag{2}$$

where $j = (-1)^{1/2}$ and $\tilde{s}_i(t)$ is the Hilbert transform of $s_i(t)$, which is obtained by

$$\tilde{s}_i(t) = \frac{1}{\pi}\mathrm{PV}\int_{-\infty}^{\infty} \frac{s_i(\tau)}{t-\tau} d\tau \tag{3}$$

where PV refers to the Cauchy principal value. The instantaneous phase $\phi_i(t)$ for channel $i$ can be computed by

$$\phi_i(t) = \arctan\frac{\tilde{s}_i(t)}{s_i(t)}. \tag{4}$$

Once the phase $\phi_i(t)(i = 1, 2, \ldots, 62)$ for all channels is obtained, the PLV and PLI between channels $i$ and $k$ can be calculated as

$$\mathrm{PLV}_{i,k} = \left|\frac{1}{T}\sum_{t=1}^{T} e^{j(\phi_i(t)-\phi_k(t))}\right| \tag{5}$$

$$\mathrm{PLI}_{i,k} = \left|\frac{1}{T}\sum_{t=1}^{T} \mathrm{sign}(\phi_i(t) - \phi_k(t))\right|. \tag{6}$$

The PLV and PLI values range from 0 to 1. No coupling is indicated with a value of 0, while perfect phase locking is indicated with a value of 1. Since each frequency band signal includes 62 channels, a connection matrix of size 62 $\times$ 62 is obtained with either the PLV or the PLI. Due to the symmetry of the connection matrix, we only collect the entries in the upper triangle. In addition, we remove the entries from the main diagonal, which represent self-connections. Finally, each band yields 1891 (62 $\times$ (62$-$1)/2) connection features, and the PLV and PLI features for a sample have 9455 dimensions (1891 connection features $\times$ 5 bands).

### D. Feature Smoothing and Standardization

According to [35], we believe that an emotional state should not be regarded as discrete but rather as a smooth trend that changes in a short time. To reduce the amount of data irrelevant to the emotional state, we assume that the emotional state changes gradually and employs a conventional moving average filter to smooth each feature sequence, a fast and robust feature smoothing method for EEG-based emotion recognition, as demonstrated in [36]. Following [15], the window length of the moving average filter is set to 20 s. For feature standardization, we independently apply the $Z$-score on the samples of each subject to mitigate the effect of intersubject variability.

### E. Feature Mapping

Because the dimensions of the FD features and BC features are of different orders of magnitude (310 versus 9455), directly concatenating these two types of features into the input for the classifier might be suboptimal. In addition, the large dimensionality of the features (310 + 9455) might lead to a high risk of overfitting [37] and might stress computational resources during subsequent training. Therefore, we map each type of feature to have the same dimensions and reduce the dimensions of the BC features. In particular, let $\mathbf{x}^{\mathrm{FD}}$ be the FD feature vector and $\mathbf{x}^{\mathrm{BC}}$ be the BC feature vector. Each type of feature vector $\mathbf{x}^m$ ($m$ = BC, FD) is mapped into the same dimensional feature vector $\mathbf{g}_m(\mathbf{x}^m) \in R^{l\times 1}$ ($m$ = FD, BC). To guarantee the quality of $\mathbf{g}_m(\mathbf{x}^m)$ ($m$ = FD, BC) with low computational costs and memory requirements, we encode each type of mapped feature $\mathbf{g}_m(\mathbf{x}^m)$ ($m$ = FD, BC) with an approximate empirical kernel map (AEKM) using the fast Nyström method [25]–[28]. The Nyström-based AEKM can extract sufficient discriminative information with small dimensions $l$, allowing for fast computation for the subsequent classifier, as demonstrated in [38] and [29].

Suppose that we are given an input feature matrix $\mathbf{X}^m = \{\mathbf{x}_i^m\}_{i=1}^n \in R^{n\times d_m}(m$ = FD, BC), the rows of which consist of $(\mathbf{x}_i^m)^{\mathrm{T}}(i = 1, 2, \ldots, n)$, where $\mathbf{x}_i^m \in R^{d_m\times 1}$ is the $i$th sample with features of type $m$ and $n$ is the number of samples. The Nyström method first finds a set of $l$ landmark points $\mathbf{V}^m = \{\mathbf{v}_j^m\}_{j=1}^l$ from $\mathbf{X}^m$, where $\mathbf{v}_j^m \in R^{d_m\times 1}$ is chosen using several selection processes, such as uniform sampling [25], column-norm sampling [26], diagonal sampling [39], or $K$-means sampling [27], [28]. We adopt a fast approximate $K$-means sampling method (i.e., Randomized Clustered Nyström method) [28] in our work; the $l$ landmark points are the cluster centroids of $\mathbf{X}^m$, which can summarize all of the training subjects, producing the generalization features

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across subjects. Next, using a kernel function $\kappa(.,.,.)$ with parameter $\sigma_m$, the Nyström method forms two matrices $\mathbf{K}_{nl}^m \in R^{n\times l}$ and $\mathbf{K}_{ll}^m \in R^{l\times l}$, where $(\mathbf{K}_{nl}^m)_{k,j} = \kappa(\mathbf{x}_k^m, \mathbf{v}_j^m, \sigma_m)$ for $k = 1,\ldots,n$ and $j = 1,\ldots,l$ and $(\mathbf{K}_{ll}^m)_{k,j} = \kappa(\mathbf{v}_k^m, \mathbf{v}_j^m, \sigma_m)$ for $k = 1,\ldots,l$ and $j = 1,\ldots,l$. Finally, with eigenvalue decomposition $\mathbf{K}_{ll}^m = \mathbf{U}_l^m \mathbf{\Lambda}_l^m \mathbf{U}_l^{m^{\mathrm{T}}}$, the AEKM $\mathbf{G}^m = \{\mathbf{g}_m(\mathbf{x}_i^m)\}_{i=1}^n \in R^{n\times l}$ for $\mathbf{X}^m$ can be obtained by

$$\mathbf{G}^m = \mathbf{K}_{nl}^m \mathbf{M}^m, \quad \text{and } \mathbf{M}^m = \mathbf{U}_l^m \left(\mathbf{\Lambda}_l^m\right)^{-\frac{1}{2}} \tag{7}$$

where $\mathbf{\Lambda}_l^m \in R^{l\times l}$ is a diagonal matrix, the diagonal entries of which are the eigenvalues of $\mathbf{K}_{ll}^m$; the columns of $\mathbf{U}_l^m \in R^{l\times l}$ are the corresponding eigenvectors; and $\mathbf{M}^m$ acts as the mapping matrix [38]. The AEKM $\mathbf{G}^m$ is described in Algorithm 1.

**Algorithm 1** : Randomized Clustered Nyström-Based AEKM for Each Type of Feature

**Input:** The input feature data $\mathbf{X}^m = \{\mathbf{x}_i^m\}_{i=1}^n$, ($m$ =*FD*, *BC*), kernel function $\kappa(.,.,.)$, and kernel parameter $\sigma_m$

**Output:** The AEKM $\mathbf{G}^m$ for input feature data $\mathbf{X}^m$

**1:** Use a fast approximate K-means sampling method [28] to find a set of $l$ landmark points $\mathbf{V}^m = \{\mathbf{v}_j^m\}_{j=1}^l$

**2:** Form the kernel matrix $\mathbf{K}_{nl}^m \in \mathrm{R}^{n\times l}$ and $\mathbf{K}_{ll}^m \in \mathrm{R}^{l\times l}$

$$\left(\mathbf{K}_{nl}^m\right)_{k,j} = \kappa\left(\mathbf{x}_k^m, \mathbf{v}_j^m, \sigma_m\right)$$

and

$$\left(\mathbf{K}_{ll}^m\right)_{k,j} = \kappa\left(\mathbf{v}_k^m, \mathbf{v}_j^m, \sigma_m\right)$$

**3:** Compute the matrices $\mathbf{U}_l^m \in \mathrm{R}^{l\times l}$ and $\mathbf{\Lambda}_l^m \in \mathrm{R}^{l\times l}$ by performing eigenvalue decomposition on $\mathbf{K}_{ll}^m$

$$\mathbf{K}_{ll}^m = \mathbf{U}_l^m \mathbf{\Lambda}_l^m \mathbf{U}_l^{m^{\mathrm{T}}}$$

**4:** Compute the mapping matrix $\mathbf{M}^m \in \mathrm{R}^{l\times l}$

$$\mathbf{M}^m = \mathbf{U}_l^m \left(\mathbf{\Lambda}_l^m\right)^{-\frac{1}{2}}$$

**5:** Compute the AEKM $\mathbf{G}^m \in \mathrm{R}^{n\times l}$

$$\mathbf{G}^m = \mathbf{K}_{nl}^m \mathbf{M}^m$$

**6: Return** AEKM $\mathbf{G}^m = \{\mathbf{g}_m(\mathbf{x}_i^m)\}_{i=1}^n$ ($m$=*FD*, *BC*)

For simplicity, the dimension $l$ is selected from a small range {200, 400, 600, 800, 1000}. The radial basis function (RBF) kernel is adopted for the AEKM and its width $\sigma_m$ is simply set to the average distance between patterns.

### *F. Classification*

SVM has been widely employed in EEG-based emotion recognition [22], [23], [40] due to its well-known generalizability. In addition, linear SVM requires the tuning of only one hyperparameter (i.e., the penalty factor), and it delivers good results under a wide range of parameter settings on all datasets (described in Section III-B), rendering the comparisons between different features fair. Therefore, in our work, SVM is adopted to test the robustness and effectiveness of the fused features in emotion recognition.

The features $\mathbf{g}_m(\mathbf{x}^m)$ ($m$ = FD, BC) mapped with the AEKM are concatenated and fed into an SVM classifier. We refer to this method as the AEKM-Fusion-SVM method below for convenience. In particular, $\mathbf{g}_{\mathrm{FD}}(\mathbf{x}^{\mathrm{FD}})$ is the mapped features of the FD features, $\mathbf{g}_{\mathrm{BC}}(\mathbf{x}^{\mathrm{BC}})$ is the mapped features of the BC features, and the fused features are denoted by $\mathbf{f}(\mathbf{x}) = [\mathbf{g}_{\mathrm{FD}}(\mathbf{x}^{\mathrm{FD}}), \mathbf{g}_{\mathrm{BC}}(\mathbf{x}^{\mathrm{BC}})]$, where $\mathbf{x} = [\mathbf{x}^{\mathrm{FD}}, \mathbf{x}^{\mathrm{BC}}]$.Given a training set $\{(\mathbf{x}_i, y_i)\}_{i=1}^n$, with $n$ samples and label $y_i \in \{-1, 1\}$, SVM solves the following optimization problem:

$$\begin{aligned} &\min_{\mathbf{w},b,\boldsymbol{\varepsilon}} \frac{1}{2}\| \mathbf{w} \|^2 + \lambda \sum_{i=1}^n \varepsilon_i \\ &\text{s.t. } y_i\left(\mathbf{w}^{\mathrm{T}}\mathbf{f}(\mathbf{x}_i) + b\right) \geq 1 - \varepsilon_i, \quad i = 1, 2, \ldots, n \end{aligned} \tag{8}$$

where $\varepsilon_i \geq 0$ denotes the tolerable error for sample $\mathbf{x}_i$ and $\lambda > 0$ is the penalty factor that governs the degree of penalty for sample misclassification. A one-against-all strategy is adopted to enable multiclass classification. We find the optimal value of $\lambda$ with the search space of $2^{[-20:20]}$.

We employ leave-one-subject-out cross validation to find the optimal parameters $l$ and $\lambda$ for the AEKM-Fusion-SVM method and assess the performance of the fused features and the single features. In particular, we choose one subject from the dataset as the test subject and leave the others as training subjects. The SEED dataset has a training set of $3394 \times 14 = 47\,516$ samples from 14 subjects and a test set of 3394 samples from one subject. For the BCI2020-A and BCI2020-B datasets, the training set consists of $(3394 \times 14) + (3370 \times 8) = 74\,476$ or $(3394 \times 15) + (3370 \times 7) = 74\,500$ samples from 22 subjects, while the test set comprises 3394 or 3370 samples from one subject.

### *G. Investigating the Properties of the Fused Features From Multiple Perspectives*

We first investigate the critical bands for cross-subject emotion recognition. Next, we introduce an index representing the percentage improvement in accuracy (PIA) to evaluate the emotional states that can be identified more accurately with the fused features compared to the single features. Finally, we study the critical channels and crucial connectivity of the fused features. The details are given as follows.

*1) Analysis of the Classification Performance of Three Emotional States:* We construct a confusion matrix to visualize the classification performance for the three emotional states based on the FD features, the BC features, and their fusion. To reach an objective conclusion, we introduce an index representing the PIA based on the confusion matrix. The PIA for each emotional state is calculated as follows:

$$\mathrm{PIA} = \frac{\mathrm{Acc}_F - \mathrm{Max}(\mathrm{Acc}_{\mathrm{FD}}, \mathrm{Acc}_{\mathrm{BC}})}{\mathrm{Max}(\mathrm{Acc}_{\mathrm{FD}}, \mathrm{Acc}_{\mathrm{BC}})} \tag{9}$$

where $\mathrm{Acc}_F$ denotes the accuracy of the fused features, $\mathrm{Acc}_{\mathrm{FD}}$ denotes the accuracy of the FD features, and $\mathrm{Acc}_{\mathrm{BC}}$ denotes the accuracy of the BC features.

*2) Critical Fused Features, Channels, and Connections:* We fuse the extracted features of each channel to investigate the crucial channels for emotion recognition. Since the BC feature

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involves two channels to calculate their correlation, we mix the single BC features with the corresponding channel-pair FD features. The fused features are referred to as the connection-channel features. In particular, let $\mathbf{x}_k^{\mathrm{BC}}$ be the BC features extracted from channels $k_1$ and $k_2$, and let $\mathbf{x}_{k_1}^{\mathrm{FD}}$ and $\mathbf{x}_{k_2}^{\mathrm{FD}}$ be the FD features of channels $k_1$ and $k_2$, respectively. We concatenate these features to obtain the connection-channel feature $\mathbf{x}_k^F = [\mathbf{x}_k^{\mathrm{BC}}, \mathbf{x}_{k_1}^{\mathrm{FD}}, \mathbf{x}_{k_2}^{\mathrm{FD}}]$. Due to symmetry, the number of connection-channel features is $C \times (C-1)/2 = 1891$, where $C$ is the number of channels (62 in the datasets).

For a set of data, $\mathbf{X}_k^{\mathrm{BC}} = \{\mathbf{x}_{i,k}^{\mathrm{BC}}\}_{i=1}^n$, $\mathbf{X}_{k_1}^{\mathrm{FD}} = \{\mathbf{x}_{i,k_1}^{\mathrm{FD}}\}_{i=1}^n$, $\mathbf{X}_{k_2}^{\mathrm{FD}} = \{\mathbf{x}_{i,k_2}^{\mathrm{FD}}\}_{i=1}^n$, $\mathbf{X}_k^F = [\mathbf{X}_k^{\mathrm{BC}}, \mathbf{X}_{k_1}^{\mathrm{FD}}, \mathbf{X}_{k_2}^{\mathrm{FD}}]$, and the corresponding labels $\mathbf{Y} = \{\mathrm{y}_i\}_{i=1}^n$, we train the SVM on the connection-channel feature matrix $\mathbf{X}_k^F$ and the labels $\mathbf{Y}$ for emotion recognition and record the accuracy using leave-one-subject-out cross validation. After sorting the accuracies from high to low, we proportionally select the top connection-channel features and then merge them to produce the important connection-channel features $\mathbf{x}_*^F$ and obtain the critical channels $\mathrm{index}_c^*$ and crucial connections $\mathrm{index}_{\mathrm{con}}^*$. The proportional threshold $t$ ranges from [0.01, 0.03] with a step size of 0.001. The method is detailed in Algorithm 2.

According to (2)–(6) and the statistics of the computation time, the computation time of the BC features is dominated by the Hilbert transform, which is applied to each channel. Therefore, in practice, we can adopt full connections based on the selected crucial channels $\mathrm{index}_c^*$ instead of the crucial connections $\mathrm{index}_{\mathrm{con}}^*$, increasing the emotion recognition accuracy. We show this adoption in Section III-C.

**Algorithm 2** : Proposed Algorithm for Critical Fused Features, Crucial Channels, and Crucial Connections

**Input:** BC feature matrix $\mathbf{X}^{BC} = \{\mathbf{x}_i^{BC}\}_{i=1}^n$, FD feature matrix $\mathbf{X}^{FD} = \{\mathbf{x}_i^{FD}\}_{i=1}^n$, labels $\mathbf{Y} = \{\mathrm{y}_i\}_{i=1}^n$, and proportional threshold $t$, and the number of channels C.

**Output:** Important connection-channel features $\mathbf{x}_*^F$, critical channels $\mathrm{index}_c^*$, and crucial connections $\mathrm{index}_{con}^*$

1: **for** $k = 1, 2, \ldots, \mathrm{C} \times (\mathrm{C}-1)/2$ **do**

2: Mix BC feature matrix $\mathbf{X}_k^{BC}$ and the corresponding channel-pair FD feature matrices $\mathbf{X}_{k_1}^{FD}$ and $\mathbf{X}_{k_2}^{FD}$, producing the connection-channel feature matrix $\mathbf{X}_k^F$

$$\mathbf{X}_k^F = \left[\mathbf{X}_k^{BC}, \mathbf{X}_{k_1}^{FD}, \mathbf{X}_{k_2}^{FD}\right]$$

3: $\mathbf{X}_k^F$ is fed into the SVM with labels $\mathbf{Y}$ using leave-one-subject-out cross-validation, and the accuracy $\mathrm{Acc}_k$ is retained in the array $\mathrm{Acc}_{set}$:

$$\mathrm{Acc}_k = \mathrm{SVM}\left(\mathbf{X}_k^F, \mathbf{Y}\right)$$
$$\mathrm{Acc}_{set} = [\mathrm{Acc}_{set}, \mathrm{Acc}_k]$$

4: **end for**

5: All accuracies are sorted from high to low, and the sorted index $\mathrm{index}_{\mathrm{F}}^*$ is obtained:

$$\mathrm{index}_{\mathrm{F}}^* = argsort(\mathrm{Acc}_{set})$$

6: Proportionally select the important connection-channel features $\mathbf{x}_*^F$ using threshold $t \in [0.01, 0.03]$:

$$\mathbf{x}_*^F = \mathbf{x}^F_{\mathrm{index}_{\mathrm{F}}^*(1:floor(t*\mathrm{C}*(\mathrm{C}-1)/2))}$$

7: Derive the critical channels $\mathrm{index}_c^*$ and crucial connections $\mathrm{index}_{con}^*$ from $\mathrm{index}_F^*$ and remove the duplicated channel indices

8: **Return** $\mathbf{x}_*^F$, $\mathrm{index}_c^*$, $\mathrm{index}_{con}^*$

## III. Experimental Results

In this section, we introduce the experimental results for the fused features for cross-subject emotion recognition. They include feature exhibition, feature classification performance, and investigations of the properties of the fused features.

### A. Feature Exhibition

We find that the feature performance of the three datasets is essentially the same; hence, we adopt the SEED dataset for preanalysis. Fig. 4(a) and (b) shows the average DE and PSD distributions for all subjects, respectively, with three activation topologies in each frequency band. These findings indicate that the neural patterns of neutral emotions are similar to those of negative emotions in the beta and gamma bands, consistent with [13]. In contrast, the positive emotional patterns have greater specificity in the activation patterns of multiple frequency bands, especially in the beta and gamma bands. It is obvious that the lateral temporal area is activated more for positive emotions than for negative emotions in the beta and gamma bands. In addition, according to Fig. 4(b) (PSD distribution), the right occipital lobe area has higher activation for the three emotional states.

Aside from the DE and PSD features, one-way analysis of variance (ANOVA) was conducted to determine whether the connectivity strength between EEG channels was significantly different among negative, positive, and neutral emotions among the five frequency bands. The significance level is set to $p < 0.01$. To facilitate observation, we select the top 1% significant connections for visualization, as shown in Fig. 4(c) and (d). The findings reveal that significant differences are present primarily in connections between nodes located in the temporal lobe, implying that the temporal lobe region contains more information about emotion recognition. In addition, as shown in Fig. 4(d), in the beta and gamma bands, apart from the connections in the temporal lobe, the frontal lobe links show more active interactions during emotional brain activities, consistent with [20].

### B. Classification Performance of AEKM-Fusion Method

*1) Influence of the Mapping Dimension:* The performances of the AEKM-Fusion-SVM method based on different features are plotted as a function of mapping dimension $l$ in Fig. 5. For comparison, the performance of the single features is also included in Fig. 5. The single features are mapped with $l$ dimensions using AEKM and then fed into the SVM classifier to quantify the performance. We observe that under different values of $l$, the classification accuracy varies very

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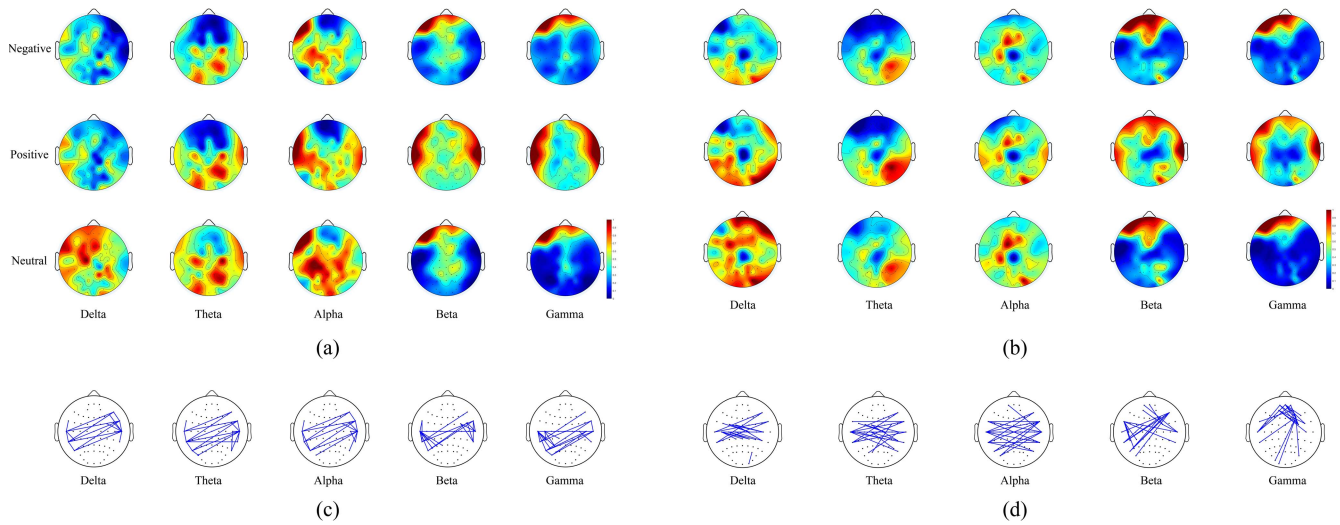



Fig. 4. Features exhibition for the SEED dataset. (a) DE distribution of the three emotional states in five frequency bands. (b) PSD distribution of the three emotional states in five frequency bands. (c) Significant differences in the PLV feature among the three emotion states in five frequency bands. (d) Significant differences in the PLI feature among the three emotion states in five frequency bands.

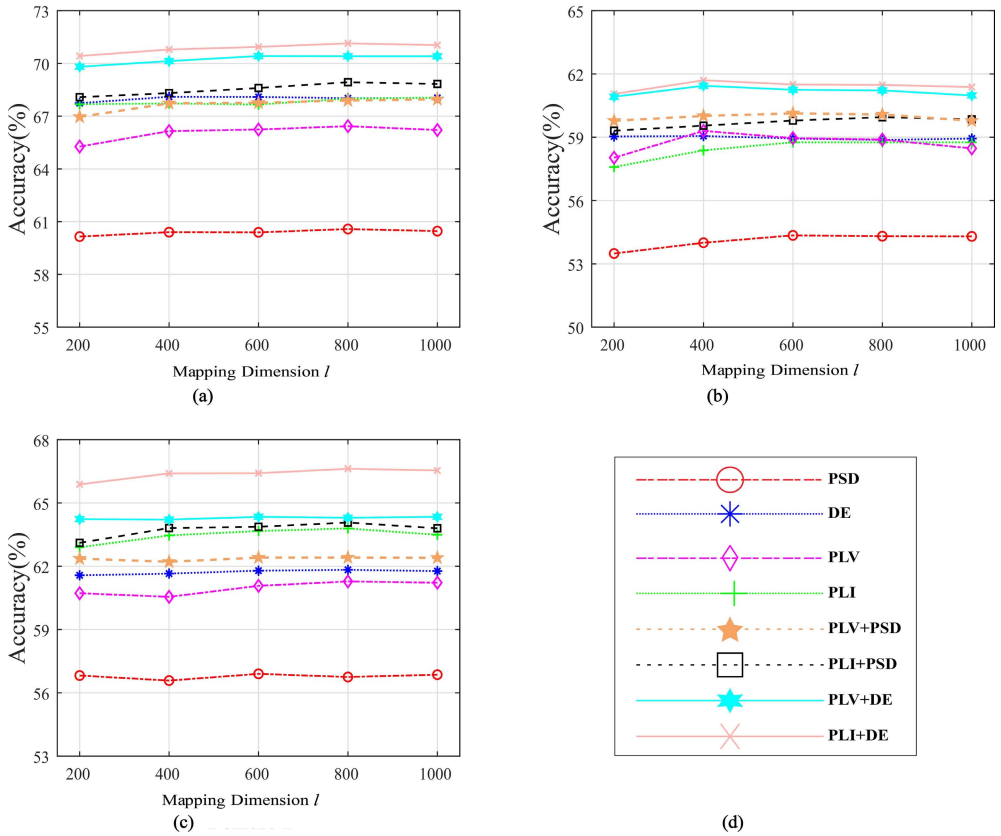



Fig. 5. Influence of the mapping dimension $l$ on the AEKM-Fusion-SVM method. (a) SEED. (b) BCI2020-A. (c) BCI2020-B. (d) Legend.

little, demonstrating that the AEKM-Fusion-SVM method is insensitive to this value. Furthermore, Fig. 5 shows that, for the three datasets, the classification accuracy of the fused features is greater than that of the corresponding single features that constitute the fused feature. These experimental results demonstrate the robustness and effectiveness of fused features in emotion recognition.

*2) Influence of the Values of Penalty Factor:* We set $l = 800$ in the AEKM-Fusion-SVM method, achieving more stable performance on all of the datasets according to Fig. 5, to investigate the influence of the penalty factor $\lambda$ in the SVM. As shown in Fig. 6, the AEKM-Fusion-SVM method remains robust across a wide range of parameter settings, and the optimal value is found among $2^{[-5,-4,-3]}$ for all fused

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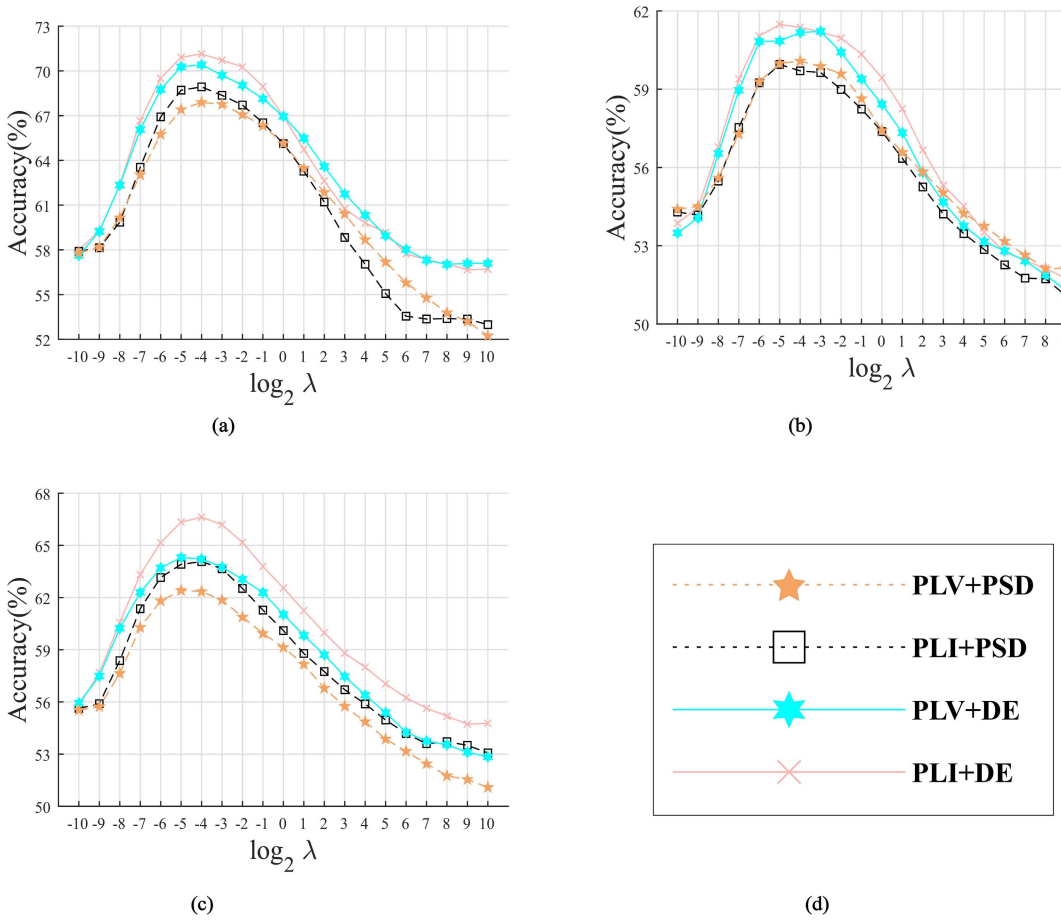



Fig. 6. Influence of the penalty factor $\lambda$ on the AEKM-Fusion-SVM method. (a) SEED. (b) BCI2020-A. (c) BCI2020-B. (d) Legend.

features on all datasets. Even though the value of $\lambda$ is fixed (e.g., $\lambda = 2^{-4}$) throughout all of the datasets, the AEKM-Fusion-SVM method still achieves satisfactory performance.

*3) Comparison With Related Fusion Methods:* We compare the performance of the direct-fusion features, canonical correlation analysis (CCA)-fusion features, DCCA-fusion features, and AEKM-fusion features in Fig. 7. The direct-fusion features are a direct concatenation of the two types of features, and CCA-fusion features and DCCA-fusion features are concatenations of the two types of features using CCA [41], [42] and DCCA [43], respectively. For CCA, the regulation parameter of autocovariance is selected from $\{10^{-8}, 10^{-7}, \ldots, 10^{-1}\}$, and the projection dimensionality is chosen from $10, 20, 30, \ldots, 300$. The projection dimensionality in CCA should be less than the dimensionality of each feature set (the dimensionality of FD features is 310 in the adopted datasets). In addition, we search the DCCA structure and its parameter space according to a recent paper on multimodal emotion recognition [23].

From Fig. 7, we have the following observations.

1) The AEKM-fusion features generalize better than the direct-fusion features. The reason might be that the dimensions of the FD features and BC features are of different orders of magnitude (310 versus 9455) and directly concatenating these two types of features into the input for the classifier cannot achieve satisfactory performance. In addition, the large dimensionality of the features (310 + 9455) might lead to a high risk of overfitting [37].
2) The CCA-fusion features perform suboptimally since CCA only calculates the linear correlation between two features, whereas the true relationship between features could be nonlinear.
3) The AEKM-fusion method achieves better performance than the DCCA-fusion method in most cases, again indicating the robustness and effectiveness of the adopted fusion methods. AEKM can produce subject-invariant features since the landmark points in AEKM are the cluster centroids of all training subjects. It is noteworthy that although DCCA-fusion features have been demonstrated to be superior in intrasubject emotion recognition [23], their performance in cross-subject emotion recognition has not been reported. Since the test subject and training subjects are sampled from different probability distributions, the optimal parameters of DCCA found for the training subjects might not generalize well to the test subject.

The time comparison for each fused feature is shown in Fig. 8 (considering the SEED dataset as an illustrative example). Compared with those of the direct-fusion features, the dimensions of the AEKM-fusion features are also greatly reduced (800 + 800 versus 310 + 9455) due to feature

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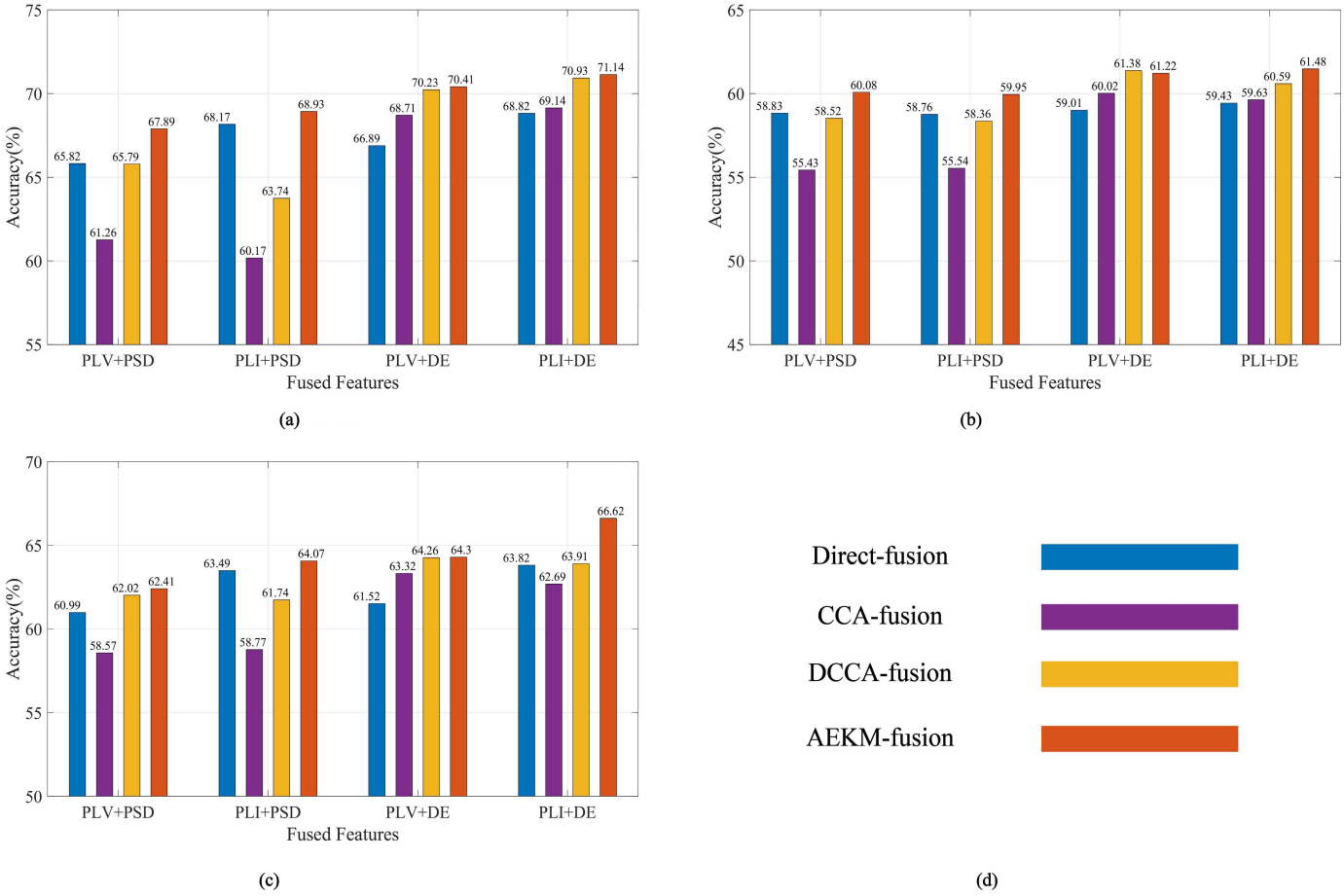



Fig. 7. Classification performance of the direct-fusion features, CCA-fusion features, DCCA-fusion features, and AEKM-fusion features. (a) SEED. (b) BCI2020-A. (c) BCI2020-B. (d) Legend.

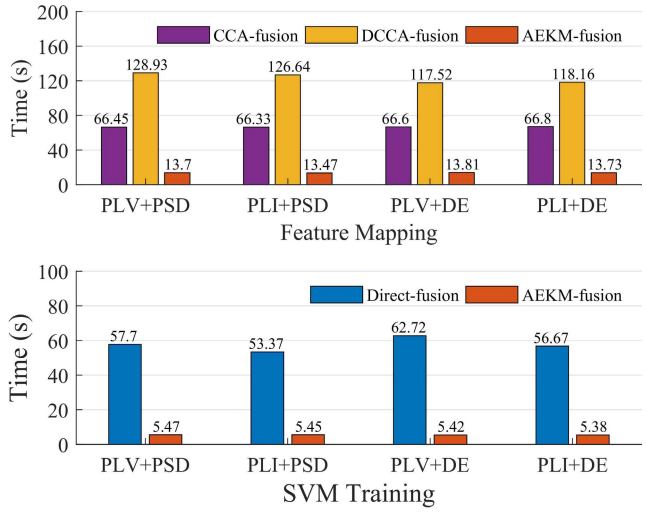



Fig. 8. Computational time for the direct-fusion features, AEKM-fusion features, CCA-fusion features, and DCCA-fusion features. The computational time is evaluated using a model with fixed parameters. The experiments were conducted using MATLAB in Windows 10 with a 2.90-GHz Intel Core i7 and 32 GB of memory. DCCA also ran with an NVIDIA GeForce RTX 3070 GPU for speedup.

mapping, substantially reducing the computational time for SVM training. Compared with those of the CCA-fusion and DCCA-fusion methods, the procedure of the AEKM-fusion method is much more computationally efficient. It is worth noting that the computational time of DCCA-fusion is not comparable to that of AEKM-fusion because DCCA-fusion contains multiple layers with many parameters and requires many backpropagations, which were conducted under GPU in the experiment. More importantly, compared with DCCA, which involves many more hyperparameters, AEKM requires the tuning of only one hyperparameter (i.e., the dimension $l$), which is simple to implement and use.

### *C. Investigating the Properties of the Fused Feature*

In the subsequent experiments, we set $l = 800$ in the AEKM-Fusion-SVM method, which achieves more stable performance on all of the datasets according to Fig. 5, to investigate the performance of the fused features. For a fair comparison, the single features are also mapped to $l$ dimensions using AEKM and then fed into the SVM classifier to report their performance.

*1) Classification Accuracy for Each Frequency Band:* The classification accuracies for each band with the three different emotion datasets are shown in Tables I–III. As seen in these tables, almost all of the fused features perform better than the corresponding single features that constitute the fused features in all five frequency bands (delta, theta, alpha, beta, and gamma); this outcome is more evident in the high-frequency

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TABLE I

AVERAGE ACCURACIES (%) FOR DIFFERENT FEATURES OBTAINED FROM SEPARATE AND ALL FREQUENCY BANDS ON THE SEED DATASET

| Features | Delta | Theta | Alpha | Beta | Gamma | All |
|---|---|---|---|---|---|---|
| PSD | 43.46±8.49 | 42.09±5.77 | 44.71±7.13 | 57.09±9.75 | 57.86±10.61 | 60.58±10.85 |
| DE | 41.64±8.48 | 42.36±7.17 | 48.35±8.93 | 63.26±7.12 | 65.05±6.85 | 68.02±6.37 |
| PLV | 60.92±8.39 | 59.14±9.41 | 56.74±9.46 | 59.69±9.10 | 59.42±6.95 | 66.43±10.86 |
| PLI | 58.90±10.79 | 61.39±10.48 | 56.24±11.92 | 62.32±7.96 | 61.82±7.74 | 68.00±10.29 |
| PLV+PSD | **60.03**±8.85 | 58.62±9.62 | 57.21±10.04 | 63.67±8.86 | 62.38±7.64 | 67.89±8.94 |
| PLI+PSD | 58.53±8.82 | **61.63**±9.45 | **57.61**±11.30 | 63.99±7.35 | 64.66±7.65 | 68.93±9.12 |
| PLV+DE | 59.75±9.36 | 58.44±9.69 | 56.89±8.55 | 65.49±7.64 | 64.80±6.37 | 70.41±7.82 |
| PLI+DE | 58.44±9.55 | 60.28±10.27 | 57.44±9.19 | **66.00**±7.08 | **66.82**±5.84 | **71.14**±6.97 |

TABLE II

AVERAGE ACCURACIES (%) FOR DIFFERENT FEATURES OBTAINED FROM SEPARATE AND ALL FREQUENCY BANDS ON THE BCI2020-A DATASET

| Features | Delta | Theta | Alpha | Beta | Gamma | All |
|---|---|---|---|---|---|---|
| PSD | 41.97±5.99 | 41.92±5.79 | 42.19±7.12 | 51.04±8.63 | 49.56±10.27 | 54.31±10.74 |
| DE | 40.39±6.69 | 39.03±5.37 | 40.74±7.44 | 56.12±8.97 | 55.85±10.85 | 58.87±11.57 |
| PLV | 54.25±10.89 | 51.37±9.84 | 49.57±10.15 | 54.61±10.86 | 55.48±7.65 | 58.89±9.83 |
| PLI | 53.70±11.21 | 53.41±13.02 | 50.95±9.97 | 54.55±11.23 | 54.36±11.36 | 58.76±11.74 |
| PLV+PSD | **54.05**±10.37 | 51.67±9.55 | 50.60±10.31 | 57.52±10.73 | 57.79±8.75 | 60.08±10.81 |
| PLI+PSD | 53.80±10.07 | **53.86**±12.64 | **51.63**±10.84 | 57.49±9.71 | 56.43±9.67 | 59.95±11.13 |
| PLV+DE | 52.92±10.71 | 50.73±10.04 | 49.77±9.89 | **59.14**±10.26 | 58.66±9.61 | 61.22±10.17 |
| PLI+DE | 53.15±10.80 | 53.10±12.38 | 50.61±10.97 | 58.98±10.52 | **59.04**±9.65 | **61.48**±10.97 |

TABLE III

AVERAGE ACCURACIES (%) FOR DIFFERENT FEATURES OBTAINED FROM SEPARATE AND ALL FREQUENCY BANDS ON THE BCI2020-B DATASET

| Features | Delta | Theta | Alpha | Beta | Gamma | All |
|---|---|---|---|---|---|---|
| PSD | 43.56±4.90 | 41.58±5.54 | 44.12±7.72 | 51.23±7.77 | 52.12±7.85 | 56.75±8.66 |
| DE | 43.97±4.97 | 41.22±5.48 | 44.46±9.11 | 57.13±8.07 | 57.96±12.08 | 61.83±10.54 |
| PLV | 55.09±10.54 | 53.65±9.59 | 57.13±9.12 | 57.19±9.45 | 56.25±7.23 | 61.28±8.68 |
| PLI | 54.07±10.11 | 56.61±9.99 | 57.32±8.67 | 58.15±8.13 | 58.07±9.27 | 63.79±9.14 |
| PLV+PSD | 55.45±9.20 | 53.86±9.78 | **58.28**±9.11 | 58.35±9.26 | 57.06±8.02 | 62.41±9.05 |
| PLI+PSD | 54.45±9.23 | **56.46**±9.38 | 57.96±8.71 | 58.60±7.37 | 58.76±8.90 | 64.07±9.05 |
| PLV+DE | **55.57**±9.36 | 53.34±9.06 | 57.59±9.05 | 59.71±9.40 | 59.41±10.50 | 64.30±9.45 |
| PLI+DE | 54.63±9.08 | 56.09±8.71 | 57.33±8.00 | **60.57**±8.20 | **62.05**±10.87 | **66.62**±9.10 |

bands (i.e., the beta and gamma bands). Meanwhile, it is observed that the fusion of PLI and DE features achieves the best classification performance in all three datasets.

*2) Investigation of Complementary Characteristics:* To further investigate the complementary characteristics of the FD features and BC features, we adopt the fused feature (i.e., PLI + DE feature) that achieved the best classification accuracy to analyze the confusion matrices. These confusion matrices demonstrate the strengths and weaknesses of every single feature in classifying different types of emotions and the stable strength of the fused feature. Fig. 9 shows the confusion graphs and the confusion matrices of the single features and fused features. The results reveal that the classification accuracy of positive emotions is the best. On the other hand, the FD and BC features have different emphases for different emotion state classifications, indicating that they have important complementary characteristics. Fig. 10 shows the PIA with the fused features for different emotion categories based on (9). We observe that the fused features achieve stable improvement for negative emotions on the three emotion datasets.

*3) Critical Fused Features, Channels, and Connections:* The three datasets were obtained with a 62-channel electrode cap. In the real world, such a large number of electrodes are not conducive to the collection of EEG data, so we explore a stable fused feature with fewer electrodes in this section. In particular, we adopt the method in Algorithm 2 to investigate the critical fused features and crucial connections and channels.

After calculation, the selected 27 electrodes and 37 connections are shown in Fig. 11. In addition, we further investigate the classification performance of the full connections of these 27 electrodes (i.e., $27 \times (27 - 1)/2 = 351$). The classification accuracies for the 27-channel (with 37 connections), 27-channel$*$ (with 351 connections), and 62-channel (with 1891 connections) configurations are shown in Table IV. The accuracy of the 27-channel$*$ fused features is the best of these three fused features. This accuracy is accompanied by a 56.45% reduction in the number of channels used, which is very meaningful for developing wearable BCI devices.

## IV. DISCUSSION

In recent years, although significant progress has been achieved in emotion recognition, the performance of fused features for cross-subject emotion recognition remains to be fully investigated. In this article, we adopted a

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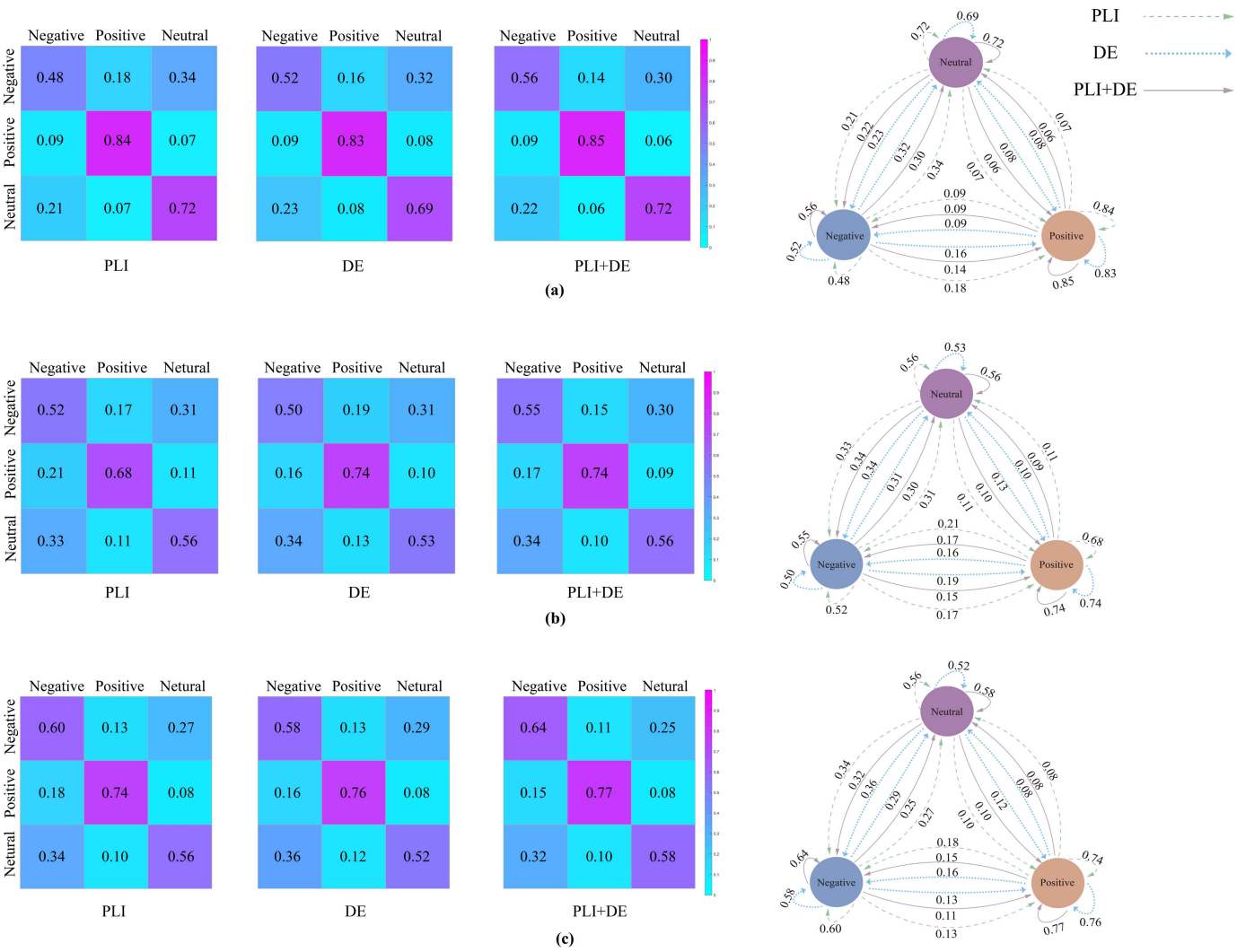



Fig. 9. Confusion matrices and confusion graphs of the single and fused features. (a) SEED dataset. (b) BCI2020-A dataset. (c) BCI2020-B dataset.

"leave-one-subject-out cross-validation" method to investigate the properties of fused features, created by merging FD features and BC features, regarding cross-subject emotion recognition.

As shown in Fig. 5, we adopted the AEKM-fusion-SVM method to investigate the classification performance of fused features in different mapping dimensions. On three emotion datasets, our adopted method demonstrated consistent robustness, effectiveness, and stability. The classification accuracies of the fused features are greater than those of the single features that constitute the fused feature. The AEKM-Fusion-SVM method also remains robust across a wide range of parameter settings, as shown in Fig. 6. In addition, we also compared the AEKM-fusion features with the direct-fusion features, CCA-fusion features, and DCCA-fusion features in terms of classification performance and computational time. In Figs. 7 and 8, we see that AEKM-fusion features not only achieve better classification accuracies but also dramatically reduce the computational time. These results demonstrate that the AEKM-fusion method generalizes well.

In our work, compared with the PSD feature, the application of the DE feature to cross-subject emotion recognition resulted in better classification performance, consistent with Zheng *et al.*'s [13] work and Li *et al.*'s [24] work. These results demonstrate that the DE feature is more suitable for EEG-based emotion recognition among the FD features. According to the results in Tables I–III, when fusing FD features with BC features, the classification accuracy improved over that of the single features in most conditions, demonstrating the importance of the complementary relationship between the FD and BC features for EEG-based cross-subject emotion recognition. Furthermore, Tables I–III present the classification performance in the delta, theta, alpha, beta, and gamma bands. The classification performance in the different frequency bands indicates that the features of the high-frequency bands (i.e., the beta and gamma bands) are more suitable for cross-subject emotion recognition than those of the other frequency bands, consistent with the results in [16] and [44]. Moreover, some studies have indicated that the high-frequency bands contain neural information related to emotions and reflect the process of emotional change [45], [46]. When we employed the fused

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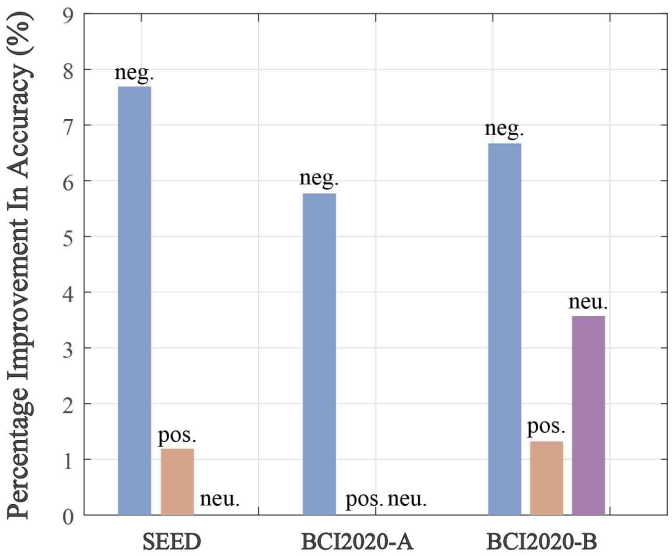



Fig. 10. Visualization of the improved accuracy of the fused features. Neg., pos., and neu. denote negative emotion, positive emotion, and neutral emotion, respectively.

features from all frequency bands, the classification accuracy was greater than that of the five individual frequency bands.

Subsequently, we chose the fused feature (i.e., PLI + DE feature) that achieved the best accuracy with the three datasets to study the classification performance regarding each emotional state for cross-subject emotion recognition. As shown by the confusion matrices and the confusion graphs of the fused feature presented in Fig. 9, the feature-fusion method stably enhances the performance in classifying emotion. In general, the results in Fig. 9 reveal that a positive emotional state can be recognized most accurately, while a negative emotional state is the most easily confused, consistent with Zheng and Lu's [16] research. We observed that, in general, the DE feature is advantageous in recognizing a positive emotion more accurately than the PLI feature, whereas the PLI feature has greater accuracy in recognizing neutral emotional states. These results indicate that the PLI and DE features have different discriminability for different emotional states, showing that the two types of features have complementary properties. In addition, we found that positive emotions and neutral emotions are most easily misclassified as negative emotions, and negative emotions are most easily recognized as neutral emotions, which could indicate that nonpositive emotions are indistinguishable in human physiological activities. In addition, we conducted a further study of the performance of the fused features in classifying the three emotions. With (9), we calculated the PIA for each emotional state. The PIAs for the negative state with the SEED, BCI2020-A, and BCI2020-B datasets are 7.69%, 5.77%, and 6.67%, respectively. The results in Fig. 10 reveal that the fused features improved upon the single features' weaknesses, significantly enhancing the classification performance for negative emotions from the three datasets. Moreover, the classification accuracies for neutral and positive emotions were also improved to a certain extent.

Finally, based on the fused features, we investigated the critical channels and crucial connections for cross-subject

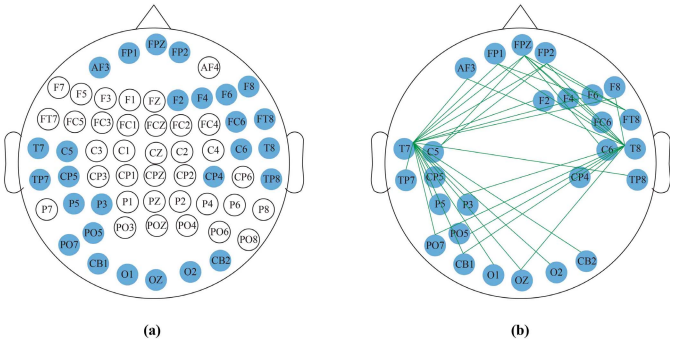



Fig. 11. Critical channels and crucial connections for the fused features: (a) selected 27 channels and (b) selected 37 connections.

TABLE IV
AVERAGE ACCURACIES (%) OF 27-CHANNEL FEATURES AND 62-CHANNEL FEATURES ON THE SEED DATASET

| Features | 27-channel (with 37 connections) | 27-channel* (with 351 connections) | 62-channel (with 1891connections) |
|---|---|---|---|
| | Acc±std | Acc±std | Acc±std |
| DE | 68.39±6.83 | 68.39±6.83 | 68.02±6.37 |
| PLI | 64.50±7.79 | 67.97±8.12 | 68.00±10.29 |
| PLI+DE | 70.09±7.29 | 71.29±6.28 | 71.14±6.97 |

Remark: Since we extracted features from 5 frequency bands, the number of features for the 27-channel (with 37 connections), 27-channel* (with 351 connections) and 62-channel (with 1891 connections) configurations are 27*5(DE) +37*5(PLI), 27*5 (DE) + 351*5 (PLI), and 62*5(DE) +1891*5(PLI), respectively.

emotion recognition. Fig. 11 shows the selected channels and connections in our work. The channels selected according to Algorithm 2 tend to be located in the frontal and lateral temporal areas. The distribution of the selected channels in our work is approximately the same as that in [13], [16], and [44]. This phenomenon can be illustrated from the following perspective: the main role of the frontal lobe is to take part in emotional control and conscious thinking activities, while the main role of the temporal lobe is to process auditory information related to emotion and memory [47]. In addition, other selected channels are located in the occipital area, which is related to visual tasks [47] and is thus responsible for processing vast amounts of visual information when the subject is watching different emotional videos. Interestingly, our 27-channel system has the same eight channels (i.e., $T7$, TP7, $C5$, CP5, FT8, $T8$, TP8, and $C6$) as Zheng and Lu's [16] 12-channel system. Then, we focused on making full use of the selected channels to establish a more accurate and practical model for classifying emotions. Therefore, we employed the full-connection PLI features of the 27 selected channels (i.e., $27 \times (27 - 1)/2 = 351$) along with the 27 channel DE features as inputs for AEKM-Fusion-SVM. Table IV compares the accuracy of the 27-channel$*$ (with 351 connections) and 62-channel (with 1891 connections) configurations based on the DE and PLI features. We can make the following observations.

1) The accuracy of the 27-channel DE features is greater than that of the 62-channel features, demonstrating that

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the selected channels are more suitable for cross-subject emotion recognition.

2) The accuracy of the 27-channel∗ PLI features (with 351 connections) is almost equal to that of the 62-channel PLI features (with 1891 connections). Considering that the number of connections employed in the 27-channel setup is reduced by 81.43%, we consider this classification accuracy to be very impressive.
3) The accuracy of the 27-channel∗ fused features (with 351 connections) is slightly better than that of the 62-channel features (with 1891 connections), accompanied by a dramatic reduction in the number of channels by 56.45%, which is more convenient for data collection in an experiment.

In summary, compared with other emotion-related studies, the contributions of this article are given as follows.

1) We directly adopted the original BC features for fusion and leveraged the AEKM-fusion method to improve the effectiveness and efficiency of the fusion of features of different dimensions. Compared with those of the direct-fusion features, the classification accuracies of the AEKM-fusion features (i.e., PLI + DE feature) on the SEED, BCI2020-A, and BCI2020-B datasets are improved by 2.32%, 2.05%, and 2.80%, respectively.

2) We applied the fused features to further improve the performance in cross-subject emotional recognition. Moreover, we discovered that the fused features stably improve the classification performance in high-frequency bands (i.e., beta and gamma).

3) We revealed that feature fusion stably increases the classification accuracy for negative emotions.

4) We developed a 27-channel system of fused features to reduce the experimental costs when applied to cross-subject emotion recognition; this change represents a reduction in the number of channels by 56.45% (relative to the original 62 channels). Interestingly, the distribution of the critical electrodes is located in the frontal, lateral temporal, and occipital areas. The centralization of critical electrodes will offer great possibilities for developing EEG wearable devices.

## V. Conclusion

In this article, by leveraging AEKM-Fusion-SVM, we systematically evaluated the performance of fused features based on FD features and BC features for cross-subject emotion recognition on three emotion datasets: SEED, BCI2020-A, and BCI2020-B. The results revealed that the AEKM-fusion method improves the effectiveness and efficiency of the fusion of features of different dimensions, and the fused features significantly outperform each individual feature, effectively improving the robustness of emotion recognition. In addition, we obtained some significant findings. First, almost all of the fused features perform significantly better than the corresponding single features in the high-frequency bands. Second, the fused features significantly improve the performance for recognizing negative emotions. Third, the fused features built with the reduced electrodes are also promising for emotion recognition.

Our study enriches the research of emotion-related brain mechanisms and provides new insight into affective computing by demonstrating that combining FD and BC features is a promising method for cross-subject emotion recognition. It is worth mentioning that we applied these fused features to the 2020 World Robot Contest-BCI Controlled Robot Contest for cross-subject emotion recognition and won second place in the finals. In future work, we will combine these fused features with transfer learning methods [48] to boost the accuracy of cross-subject emotion recognition.

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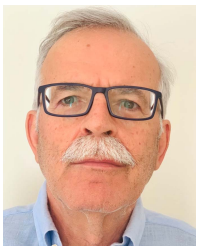
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